

linearity of $f \rightarrow F(u^i - u^j)$

$$\Leftrightarrow \frac{\|F(u^i) - F(u^j)\|^2}{\|u^i - u^j\|^2} - 1 \leq \delta$$

$\sim \mathbb{E}(X)$

$$F(v) = \frac{1}{\sqrt{m}} A v$$

$$X' = \frac{\|F(v)\|^2}{\|v\|^2} = \frac{1}{m} \frac{\|Av\|^2}{\|v\|^2}$$

$$Av = \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix} \begin{bmatrix} v \\ \vdots \\ v \end{bmatrix} = \begin{bmatrix} \langle a_1, v \rangle \\ \vdots \\ \langle a_m, v \rangle \end{bmatrix}$$

$$\frac{\|Av\|^2}{\|v\|^2} = \sum_{i=1}^m \underbrace{\left\langle \underbrace{a_i}_{\sim \mathcal{N}(0, I)}, \underbrace{\frac{v}{\|v\|}}_{\sim \mathcal{N}(0, 1)} \right\rangle^2}_{\|w\|=1}$$

$$Z \sim \mathcal{N}(\mu, \Sigma) \quad w^T Z \sim \mathcal{N}(w^T \mu, w^T \Sigma w)$$

$$\underbrace{\langle a_i, \frac{v}{\|v\|} \rangle}_{\sim Z} \sim \mathcal{N}(0, \underbrace{w^T I w}_{= \|w\|^2 = 1})$$

$$\Rightarrow \frac{\|Av\|^2}{\|v\|^2} \sim \chi_m^2$$

$$\frac{\|F(v)\|^2}{\|v\|^2} = \frac{1}{m} \frac{\|Av\|^2}{\|v\|^2}$$

$$\rightarrow \mathbb{P} \left(\left| \frac{1}{m} Y - 1 \right| \geq \delta \right) \leq 2 e^{-m \delta^2 / 8}$$

$$Y \sim \chi_m^2$$

$$\forall v \in \mathbb{R}^d \quad \mathbb{P} \left(\left| \frac{\|F(v)\|^2}{\|v\|^2} - 1 \right| \geq \delta \right) \leq 2 e^{-m \delta^2 / 8}$$

$$\mathbb{P} \left(\forall u^i, u^j \quad u^i \neq u^j \quad \left| \frac{\|F(u^i - u^j)\|^2}{\|u^i - u^j\|^2} - 1 \right| \leq \delta \right)$$

$$= 1 - \mathbb{P} \left(\exists u^i, u^j : u^i \neq u^j \quad \left| \frac{\|F(u^i - u^j)\|^2}{\|u^i - u^j\|^2} - 1 \right| > \delta \right)$$

$$= 1 - \mathbb{P} \left(\bigcup_{\substack{\{i,j\} \\ i \neq j}} \underbrace{\left| \frac{\|F(u^i - u^j)\|^2}{\|u^i - u^j\|^2} - 1 \right|}_{A_i} > \delta \right)$$

$$\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$$

$$\geq 1 - \sum_{i,j} \mathbb{P} \left(\underbrace{\left| \frac{\|F(u^i - u^j)\|^2}{\|u^i - u^j\|^2} - 1 \right|}_{\leq e^{-m\delta^2/8}} > \delta \right)$$

$$\geq 1 - 2 \binom{N}{2} e^{-m\delta^2/8}$$

$$1 - 2 \binom{N}{2} e^{-m\delta^2/8} \geq 1 - \epsilon$$

$$\ln N(N-1) - m\delta^2/8 \leq \ln \epsilon$$

$$m\delta^2/8 \geq \ln \frac{N(N-1)}{\epsilon} \Rightarrow m \geq \frac{8 \ln \frac{N(N-1)}{\epsilon}}{\delta^2}$$

Suff. to take

$$m \geq \frac{16 \ln \frac{N}{\epsilon}}{\delta^2}$$

$$\mathbb{P} \left(\underbrace{f(X_1, \dots, X_n) - \mathbb{E}(f(X_1, \dots, X_n))}_{\sum_{i=1}^n \Delta_i} \geq t \right)$$

$$\begin{aligned} \mathbb{E} \left\{ e^{\lambda (\sum \Delta_i)} \right\} &= \mathbb{E} \left\{ e^{\lambda \Delta_1} \cdot e^{\lambda \Delta_2} \cdots e^{\lambda \Delta_n} \right\} \\ &= \mathbb{E} \{ e^{\lambda \Delta_1} \} \cdot \mathbb{E} \{ e^{\lambda \Delta_2} \} \cdots \mathbb{E} \{ e^{\lambda \Delta_n} \} \\ &\quad \text{if } \Delta_i \text{'s are indep} \end{aligned}$$

$$\underline{X} = (X_1, \dots, X_n)$$

$$Y_k = \mathbb{E}(f(\underline{X}) \mid X_1, \dots, X_k)$$

$$Y_0 = \mathbb{E}(f(\underline{X}))$$

$$Y_n = f(\underline{X})$$

$$\begin{aligned} f(\underline{X}) - \mathbb{E}(f(\underline{X})) &= Y_n - Y_0 \\ &= \cancel{Y_n - Y_{n-1}} + \cancel{Y_{n-1} - Y_{n-2}} + \cdots \\ &\quad + Y_1 - Y_0 \end{aligned}$$

$$= \sum_{k=1}^n \underbrace{Y_k - Y_{k-1}}_{D_k}$$

Are D_k 's independent ?!

$$D_2 = \underbrace{Y_2 - Y_1}_{g_1(x_2, x_1)} \quad k=2$$

$$D_3 = \underbrace{Y_3 - Y_2}_{g_2(x_1, x_2, x_3)} \quad k=3$$

Martingale Seq.

σ -field

$$\mathcal{F}_i = \sigma(X_1, \dots, X_i)$$

$\mathcal{F}_1 \dots \mathcal{F}_k \dots$

$$\mathcal{F}_i \subseteq \mathcal{F}_{i+1}$$

filtration

$$\begin{cases} \mathbb{E}\{Y_i \mid \mathcal{F}_{i-1}\} = Y_{i-1} \\ \mathbb{E}\{|Y_i|\} < \infty \end{cases}$$

$(Y_k, \mathcal{F}_k)_{k=1}^{\infty}$ is called
a Martingale

$(Y_i - Y_{i-1}, \mathcal{F}_i)$ a Mart.
Diff. Seq.